

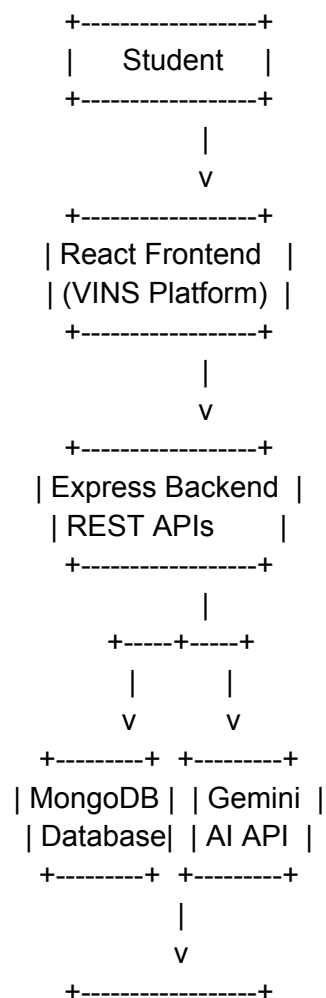
VINS – AI Internship Intelligence Platform

System Design / Architecture + Diagrams

1. System Overview

VINS (AI Internship Intelligence Platform) is an AI-powered internship support ecosystem developed using the MERN stack and Google Gemini AI. The platform helps interns find answers to frequently asked questions using a Retrieval-Augmented Generation (RAG) pipeline while reducing mentor workload through intelligent query handling and continuous learning.

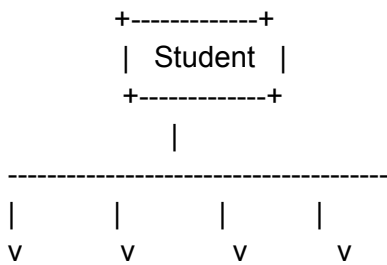
2. High-Level System Architecture



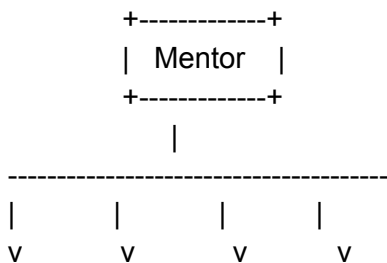
Description

- Frontend built with React.js and Tailwind CSS.
 - Backend developed using Node.js and Express.js.
 - MongoDB stores users, FAQs, queries, and analytics data.
 - Gemini AI generates contextual responses.
 - Analytics Engine tracks confusion trends and platform usage.
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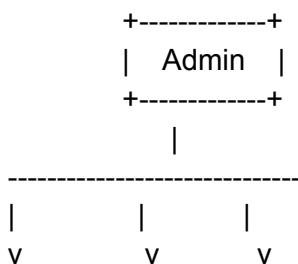
3. Use Case Diagram



Ask AI Raise Query View FAQ View Profile



Answer Add to KB View Gaps Analytics

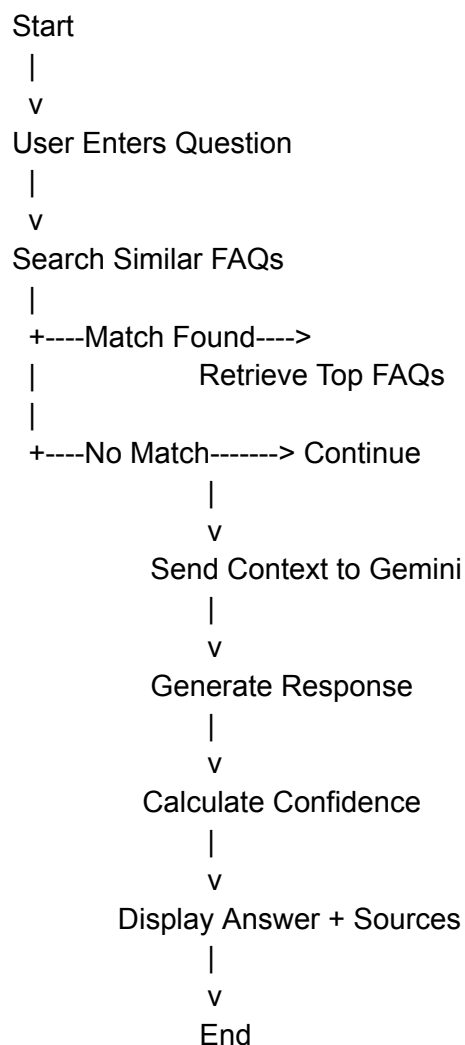


Manage Users Announcements Delete Content

Actors

1. Student
 - Ask AI questions
 - Raise queries
 - Browse FAQs
 - View profile
 2. Mentor
 - Answer queries
 - Add answers to knowledge base
 - View FAQ gaps
 - Monitor analytics
 3. Admin
 - Manage users
 - Publish announcements
 - Moderate content
-

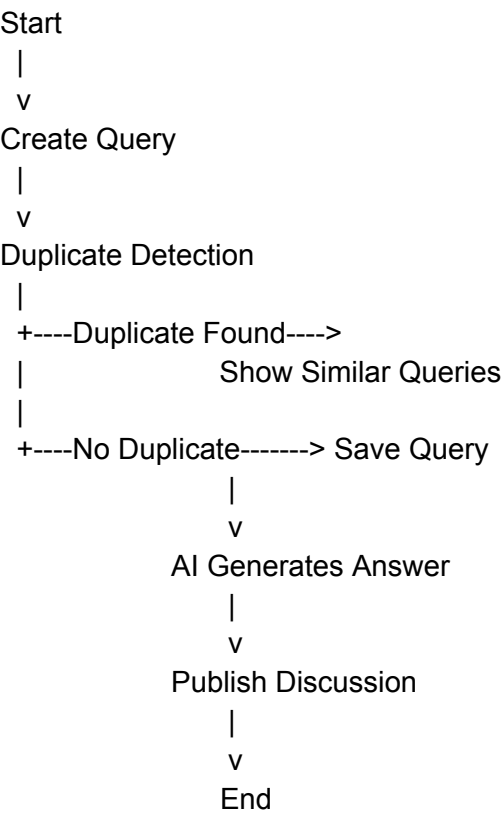
4. AI Query Processing Flowchart



Description

The system retrieves relevant FAQs, sends only validated context to Gemini AI, generates an answer, calculates confidence, and displays citations.

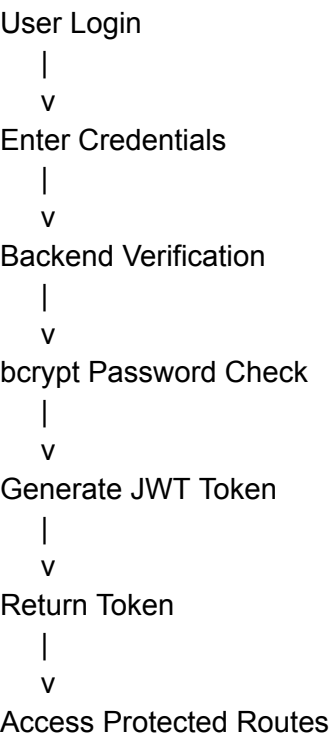
5. Query Posting Flowchart



Description

Before saving a query, the system checks for duplicates using similarity search and existing FAQs.

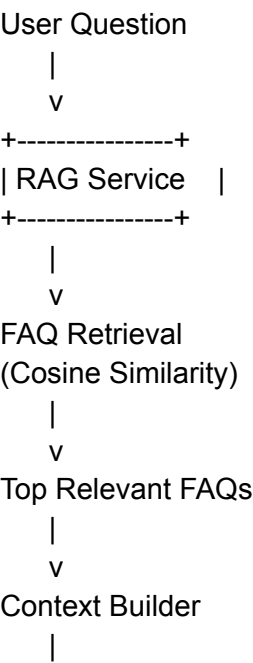
6. Authentication Flow

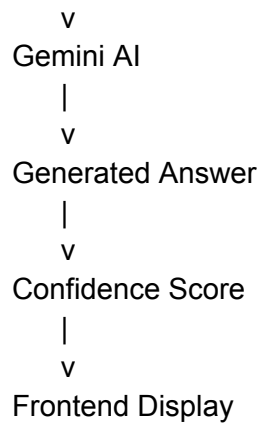


Description

JWT authentication ensures secure access to protected resources based on user roles.

7. Data Flow Diagram (DFD)

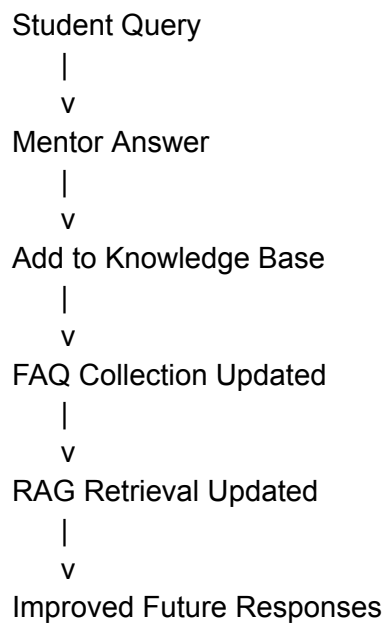




Description

The RAG pipeline minimizes hallucinations by grounding AI responses using verified FAQ data.

8. Continuous Learning Data Flow



Description

The platform continuously improves as mentor-validated answers become part of the knowledge base.

9. Database ER Diagram

```
+-----+
| User  |
+-----+
|_id    |
| name  |
| email |
| role  |
+-----+

      |
      | 1:M
      v
```

```
+-----+
| Query |
+-----+
| title |
| content|
| userId|
+-----+

      |
      | 1:M
      v
```

```
+-----+
| Answer|
+-----+
| content|
| userId|
+-----+
```

```
+-----+
| FAQ   |
+-----+
| question|
| answer |
| category|
+-----+
```

```
+-----+
| Analytics |
+-----+
| category  |
| confidence|
| date      |
+-----+
```

Description

The database stores users, FAQs, queries, answers, and analytics data required by the platform.

10. Conclusion

VINS integrates React.js, Node.js, MongoDB, and Gemini AI to provide a scalable AI-powered internship intelligence platform. Through its RAG-based architecture, confidence scoring, duplicate detection, analytics dashboard, and continuous learning mechanism, the platform reduces confusion, improves answer accuracy, and enhances the internship experience for students and mentors.